IMO 2004 Solution Notes

Evan Chen《陳誼廷》

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This is a compilation of solutions for the 2004 IMO. The ideas of the solution are a mix of my own work, the solutions provided by the competition organizers, and solutions found by the community. However, all the writing is maintained by me.

These notes will tend to be a bit more advanced and terse than the "official" solutions from the organizers. In particular, if a theorem or technique is not known to beginners but is still considered "standard", then I often prefer to use this theory anyways, rather than try to work around or conceal it. For example, in geometry problems I typically use directed angles without further comment, rather than awkwardly work around configuration issues. Similarly, sentences like "let $\mathbb{R}$ denote the set of real numbers" are typically omitted entirely.

Corrections and comments are welcome!

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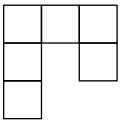
§0 Problems

1. Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC. The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R. Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC.

2. Find all polynomials P with real coefficients such that for all reals a, b, c such that $ab + bc + ca = 0$, we have

$$P(a-b) + P(b-c) + P(c-a) = 2P(a+b+c).$$

3. Define a "hook" to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotations and reflections to this figure.



Which $m \times n$ rectangles can be tiled by hooks?

4. Let $n \geq 3$ be an integer and t_1, t_2, $\ldots$, t_n positive real numbers such that

$$n^2 + 1 > (t_1 + t_2 + \cdots + t_n)\left(\frac{1}{t_1} + \frac{1}{t_2} + \cdots + \frac{1}{t_n}\right).$$

Show that t_i, t_j, t_k are the sides of a triangle for all i, j, k with $1 \leq i < j < k \leq n$.

5. In a convex quadrilateral $ABCD$, the diagonal BD bisects neither the angle ABC nor the angle CDA. The point P lies inside $ABCD$ and satisfies

$$\angle PBC = \angle DBA \quad \text{and} \quad \angle PDC = \angle BDA.$$

Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.

6. We call a positive integer *alternating* if every two consecutive digits in its decimal representation are of different parity. Find all positive integers n which have an alternating multiple.

§1 Solutions to Day 1

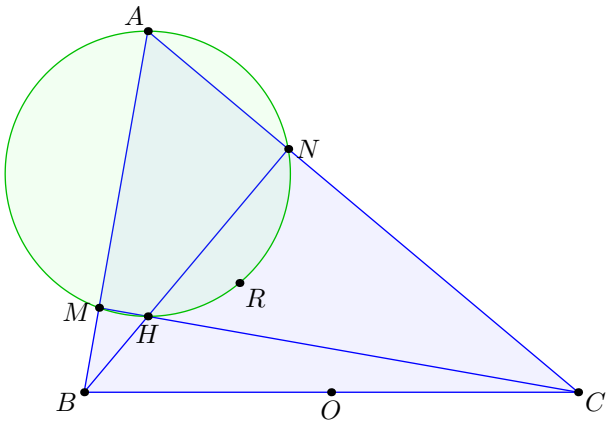
§1.1 IMO 2004/1, proposed by Dinu Serbanescu and Valentin Vornicu (ROU)

🔗 *Available online at* https://aops.com/community/p99445.

Problem statement

Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC. The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R. Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC.

By Miquel's theorem it's enough to show $AMRN$ is cyclic.



In fact, since the bisector of $\angle MON$ is just the perpendicular bisector of $\overline{MN}$, the point R is actually just the arc midpoint of $\widehat{MN}$ of (AMN) as desired.

§1.2 IMO 2004/2, proposed by Hojoo Lee (KOR)

🔗 *Available online at* https://aops.com/community/p99448.

Problem statement

Find all polynomials P with real coefficients such that for all reals a, b, c such that $ab+bc+ca=0$, we have

$$P(a-b)+P(b-c)+P(c-a)=2P(a+b+c).$$

The answer is

$$P(x)=\alpha x^4+\beta x^2$$

which can be checked to work, for any real numbers α and β.

It is easy to obtain that P is even and $P(0)=0$. The trick is now to choose $(a,b,c)=(6x,3x,-2x)$ and then compare the leading coefficients to get

$$3^n+5^n+8^n=2\cdot 7^n$$

for $n=\deg f$ (which is even). As $n\geq 7 \implies (8/7)^n>2$, this means that we must have $n\leq 6$. Moreover, taking modulo 7 we have $3^n+5^n\equiv 6 \pmod 7$ which gives $n\equiv 2,4 \pmod 6$.

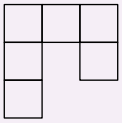
Thus $\deg P\leq 4$, which (combined with P even) resolves the problem.

§1.3 IMO 2004/3, proposed by Jan Willemson (EST)

🔗 *Available online at* https://aops.com/community/p99450.

Problem statement

Define a "hook" to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotations and reflections to this figure.



Which $m \times n$ rectangles can be tiled by hooks?

The answer is that one requires:

- $\{1,2,5\} \notin \{m,n\}$,
- $3 \mid m$ or $3 \mid n$,
- $4 \mid m$ or $4 \mid n$.

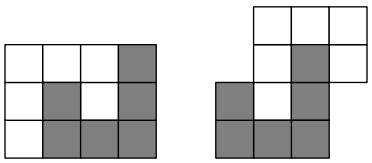
First, we check all of these work, in fact we claim:

Claim — Any rectangle satisfying these conditions can be tiled by 3×4 rectangles (and hence by hooks).

Proof. If $3 \mid m$ and $4 \mid n$, this is clear. Else suppose $12 \mid m$ but $3 \nmid n$, $4 \nmid n$. Then $n \geq 7$, so it can be written in the form $3a + 4b$ for nonnegative integers a and b, which permits a tiling. □

We now prove these conditions are necessary. It is not hard to see that $m, n \notin \{1,2,5\}$ is necessary.

We thus turn our attention to divisibility conditions. Each hook has a *hole*, and if we associate each hook with the one that fills its hole, we get a bijective pairing of hooks. Thus the number of cells is divisible by 12, and the cells come grouped into two possible shapes, which we will call **tiles** shown below, (rotations and reflections permitted).



In particular, the total number of cells is divisible by 12. Thus the problem is reduced to proving that:

Claim — if a $6a \times 2b$ rectangle is tiled by tiles, then at least one of a and b is even.

Proof. Note that the tiles can be further classified into two types:

- **First type**: These tiles have exactly four columns, each with exactly three cells (an odd number). Moreover, all rows have an even number of cells (either 2 or 4).

- **Second type**: vice-versa. These tiles have exactly four rows, each with exactly three cells (an odd number). Moreover, all rows have an odd number of cells.

We claim that any tiling uses an even number of each type, which is enough.

By symmetry we prove an even number of first-type tiles. Color red every fourth column of the rectangle. The number of cells colored red is even. The tiles of the second type cover an even number of red cells, and the tiles of the first type cover an odd number of red cells. Hence the number of tiles of the first type must be even. □

> **Remark.** This shows that a rectangle can be tiled by hooks if and only if it can be tiled by 3×4 rectangles. But there exist tilings which do not decompose into 3×4; see e.g. https://aops.com/community/c6h14023p99881.

§2 Solutions to Day 2

§2.1 IMO 2004/4, proposed by Hojoo Lee (KOR)

🔗 *Available online at* https://aops.com/community/p99756.

Problem statement

Let $n \geq 3$ be an integer and t_1, t_2, ..., t_n positive real numbers such that

$$n^2 + 1 > (t_1 + t_2 + \cdots + t_n)\left(\frac{1}{t_1} + \frac{1}{t_2} + \cdots + \frac{1}{t_n}\right).$$

Show that t_i, t_j, t_k are the sides of a triangle for all i, j, k with $1 \leq i < j < k \leq n$.

Let $a = t_1$, $b = t_2$, $c = t_3$. Expand:

$$\begin{aligned}
n^2 + 1 &> (t_1 + t_2 + \cdots + t_n)\left(\frac{1}{t_1} + \cdots + \frac{1}{t_n}\right) \\
&= n + \sum_{1 \leq i < j \leq n} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i}\right) \\
&= n + \sum_{1 \leq i < j \leq n} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i}\right) \\
&\geq n + \sum_{1 \leq i < j \leq 3} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i}\right) + \sum_{\substack{1 \leq i < j \leq n \\ j > 3}} 2 \\
&= n + 2\left(\binom{n}{2} - 3\right) + \left(\frac{a}{b} + \frac{b}{a}\right) + \frac{a+b}{c} + \frac{c}{b} + \frac{c}{a} \\
&\geq n + 2\left(\binom{n}{2} - 3\right) + 2 + \frac{a+b}{c} + c \cdot \frac{4}{a+b}
\end{aligned}$$

So, we conclude that

$$\frac{a+b}{c} + \frac{4c}{a+b} < 5$$

which rearranges to

$$(4c - (a+b))\,(c - (a+b)) < 0.$$

This is enough to imply $c < a + b$.

Remark. A variant of the same argument allows one to improve the left-hand side to $(n + \sqrt{10} - 3)^2$. One does so by writing

$$\text{RHS} \geq \left(\sqrt{(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)} + (n-3)\right)^2$$

and estimating the square root as in the previous approach.

In addition, $(n + \sqrt{10} - 3)^2$ is best possible, as seen by taking $(a, b, c) = (2, 1, 1)$ and $t_4 = t_5 = \cdots = \frac{2}{5}\sqrt{10}$.

§2.2 IMO 2004/5, proposed by Waldemar Pompe (POL)

🔗 *Available online at* https://aops.com/community/p99759.

Problem statement

In a convex quadrilateral $ABCD$, the diagonal BD bisects neither the angle ABC nor the angle CDA. The point P lies inside $ABCD$ and satisfies

$$\angle PBC = \angle DBA \quad \text{and} \quad \angle PDC = \angle BDA.$$

Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.

We show two solutions. We note that the first hypothesis cannot be dropped, because $ABCD$ being a kite with $BA = BC$ and $DA = DC$ with center P is a counterexample (where $AP = CP$ but $ABCD$ is not cyclic). However, the condition that BD bisects neither angle $\angle B$ nor $\angle D$ is equivalent to requiring that the point P does not lie on line BD and we assume this henceforth.

¶ Barycentric solution (from teenager Evan). Apply barycentric coordinates to $\triangle PBD$ (nondegenerate by the non-bisect condition) with $P = (1, 0, 0)$, $B = (0, 1, 0)$ and $D = (0, 0, 1)$. Define $a = BD$, $b = DP$ and $c = PB$.

Since A and C are isogonal conjugates with respect to $\triangle PBD$, we set

$$A = (au : bv : cw) \qquad \text{and} \qquad C = \left(\frac{a}{u} : \frac{b}{v} : \frac{c}{w} \right).$$

For brevity define $M = au + bv + cw$ and $N = avw + bwu + cuv$.

We now compute each condition.

Claim — Quadrilateral $ABCD$ is cyclic if and only if $N^2 = u^2 M^2$.

Proof. W know a circle through B and D is a locus of points with

$$\frac{a^2yz + b^2zx + c^2xy}{x(x + y + z)}$$

is equal to some constant. Therefore quadrilateral $ABCD$ is cyclic if and only if $\frac{abc \cdot N}{au \cdot M}$ is equal to $\frac{abc \cdot uvw \cdot M}{avw \cdot N}$ which rearranges to $N^2 = u^2 M^2$. □

Claim — We have $PA = PC$ if and only if $N^2 = u^2 M^2$.

Proof. We have the displacement vector $\overrightarrow{PA} = \frac{1}{M}(bv + cw, -bv, -cw)$. Therefore,

$$\begin{aligned} M^2 \cdot |PA|^2 &= -a^2(bv)(cw) + b^2(cw)(bv + cw) + c^2(bv)(bv + cw) \\ &= bc(-a^2vw + (bw + cv)(bv + cw)). \end{aligned}$$

In a similar way (by replacing u, v, w with their inverses) we have

$$\begin{aligned} &\left(\frac{N}{uvw} \right)^2 \cdot |PC|^2 = (vw)^{-2} \cdot bc(-a^2vw + (bv + cw)(bw + cv)) \\ \iff &N^2 \cdot |PC|^2 = u^2 bc(-a^2vw + (bw + cv)(bv + cw)) \end{aligned}$$

These are equal if and only if $N^2 = u^2 M^2$, as desired. □

¶ Angle chasing solution (Evan Chen and Petko Lazarov). The solution consists of two parts. The first part is that by angle chasing, we will prove that

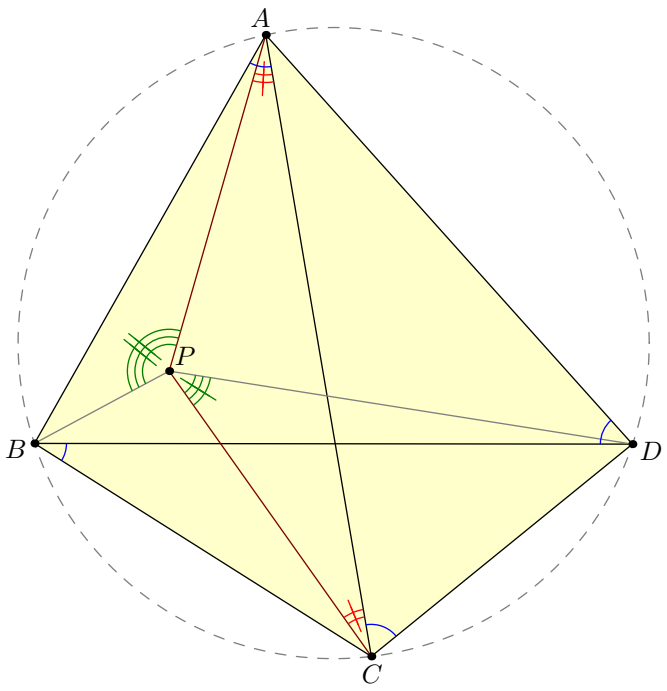
$$\measuredangle PAC = \measuredangle ACP \iff \measuredangle BAC + \measuredangle CBD + \measuredangle DCA + \measuredangle ADB = 0. \qquad (\spadesuit).$$

A careless reader would be forgiven for thinking that ($\spadesuit$) implies the problem or at least one direction, but it turns out the situation is more subtle. The second part analyzes the angle conditions more carefully and provides a complete proof.

Proof of the equivalence ($\spadesuit$) by angle chasing. We start with the following unconditional claim, valid for any quadrilateral.

> **Claim** (Isogonal conjugation) — Let $ABCD$ and P be as in the problem statement. Then $\measuredangle APB + \measuredangle CPD = 180°$.

Proof. The angles in the statement imply that A and C are isogonal conjugates with respect to $\triangle PBC$. Thus, lines PA and PC are isogonal with respect to $\angle BPC$, as needed. □



Next we rewrite the two angles $\measuredangle APB$ and $\measuredangle CPD$ in the claim (colored green with three rings) so that their only dependence on P is through the angles $\measuredangle PAC$ and $\measuredangle CAP$ (colored red with two rings), as follows:

$$\begin{aligned} -\measuredangle APB &= \measuredangle PBA + \measuredangle BAP = \measuredangle PBA + (\measuredangle BAC - \measuredangle PAC) \\ &= \measuredangle CBD + \measuredangle BAC - \measuredangle PAC \\ -\measuredangle CPD &= \measuredangle PDC + \measuredangle DCP = \measuredangle PDC + (\measuredangle DCA + \measuredangle ACP) \\ &= \measuredangle ADB + \measuredangle DCA + \measuredangle ACP. \end{aligned}$$

Since the claim says $\measuredangle APB + \measuredangle CPD = 0$, summing lets us finally rewrite $\measuredangle PAC - \measuredangle APC$ in terms of only A, B, C, D:

$$0 = (\measuredangle ACP - \measuredangle PAC) + \measuredangle ADB + \measuredangle DCA + \measuredangle CBD + \measuredangle BAC$$

$$\implies \measuredangle PAC - \measuredangle ACP = \measuredangle ADB + \measuredangle DCA + \measuredangle CBD + \measuredangle BAC.$$

These four latter angles are colored blue with one ring in the figure. This proves (♠).

Quasi-harmonic quadrilaterals. To interpret the condition (♠), we define a new term: a quadrilateral $ABCD$ is *quasi-harmonic* if $AB \cdot CD = BC \cdot DA$. (See IMO 2018/6 for another problem involving quasi-harmonic quadrilaterals.) The following two lemmas show why this condition is relevant:

Lemma

The condition
$$\measuredangle BAC + \measuredangle CBD + \measuredangle DCA + \measuredangle ADB = 0$$
is equivalent to $ABCD$ being *either* cyclic or quasi-harmonic or both.

Proof. A clean approach with complex numbers (posted at https://aops.com/community/p15327734) goes as follows: define complex numbers $\mu = (a-b)(c-d) \neq 0$ and $\nu = (b-c)(d-a) \neq 0$. Then
$$\mu - \nu = (a-c)(b-d) \neq 0.$$

Now the angle condition is equivalent to
$$\begin{aligned} \mathbb{R} \ni \frac{a-b}{d-b} \cdot \frac{b-c}{a-c} \cdot \frac{c-d}{b-d} \cdot \frac{d-a}{c-a} &= \frac{\mu\nu}{(\mu-\nu)^2} \\ &= \frac{1}{\frac{\mu}{\nu} + \frac{\nu}{\mu} - 2} \\ \iff \mathbb{R} \ni \frac{\mu}{\nu} + \frac{\nu}{\mu}. & \end{aligned}$$

In general, though $z + 1/z$ is real exactly when either $|z| = 1$ or $z \in \mathbb{R}$. Note $\mu/\nu \in \mathbb{R}$ is equivalent to $ABCD$ cyclic and $|\mu| = |\nu|$ is equivalent to $ABCD$ being quasiharmonic; the proof follows.

This can be also proved by inversion at A, though we don't write the details here. (See also https://problems.ru/view_problem_details_new.php?id=116602). □

Lemma

Quadrilateral $ABCD$ is quasi-harmonic if and only if P lies on line AC.

Proof. Let $X = \overline{BD} \cap \overline{AC}$. If the isogonal of line BD with respect to $\angle B$ meets line AC at Y, then $\frac{AX}{CX}\frac{AY}{CY} = \left(\frac{BA}{BC}\right)^2$. Similarly if the isogonal to $\angle D$ meets line AC at Y', then $\frac{AX}{CX}\frac{AY'}{CY'} = \left(\frac{DA}{DC}\right)^2$. Hence $ABCD$ is quasi-harmonic if and only if $Y = Y'$ (that is, $Y = Y' = P$). □

Wrap-up. We now show that $PA = PC$ if and only if $ABCD$ is cyclic by cases on whether P lies on $\overline{AC}$.

- If $ABCD$ is *not* quasi-harmonic, then (♠) implies the problem statement immediately. Indeed, $\measuredangle PAC = \measuredangle ACP$ if and only if $PA = PC$ (as $\triangle PAC$ is not degenerate) and the second lemma turns our angle condition into $ABCD$ cyclic.

- Now assume $ABCD$ is quasi-harmonic and P lies on line AC. We ignore (♠). Instead, note that if $ABCD$ is also cyclic then $\overline{BD}$ is a symmedian of $\triangle ABC$ and hence P is the midpoint. Conversely, suppose we know $\overline{BD}$ is a symmedian of $\triangle ABC$. Let $D' \neq B$ be the point for which $ABCD'$ is cyclic and harmonic; then B, D, D' are collinear and $\frac{BD}{CD} = \frac{BD'}{CD'} = \frac{BA}{CA}$. So $D' = D$ (the corresponding Apollonian circle only meets line BD twice), as needed.

§2.3 IMO 2004/6, proposed by Mohsen Jamaali and Armin Morabbi (IRN)

🔗 *Available online at* https://aops.com/community/p99760.

Problem statement

We call a positive integer *alternating* if every two consecutive digits in its decimal representation are of different parity. Find all positive integers n which have an alternating multiple.

If $20 \mid n$, then clearly n has no alternating multiple since the last two digits are both even. We will show the other values of n all work.

The construction is just rush-down do-it. The meat of the solution is the two following steps.

Claim (Tail construction) — For every even integer $w \geq 2$,

- there exists an even alternating multiple $g(w)$ of 2^{w+1} with exactly w digits, and
- there exists an even alternating multiple $h(w)$ of 5^w with exactly w digits.

(One might note this claim is implied by the problem, too.)

Proof. We prove the first point by induction on w. If $w = 2$, take $g(2) = 32$. In general, we can construct $g(w+2)$ from $g(w)$ by adding some element in

$$10^w \cdot \{10, 12, 14, 16, 18, 30, \ldots, 98\}$$

to $g(w)$, corresponding to the digits we want to append to the start. This multiple is automatically divisible by 2^{w+1}, and also can take any of the four possible values modulo 2^{w+3}.

The second point is a similar induction, with base case $h(2) = 50$. The same set above consists of numbers divisible by 5^w, and covers all residues modulo 5^{w+2}. Careful readers might recognize the second part as essentially USAMO 2003/1. □

Claim (Head construction) — If $\gcd(n, 10) = 1$, then for any b, there exists an even alternating number $f(b \bmod n)$ which is $b \pmod n$.

Proof. A standard argument shows that

$$10 \cdot \frac{100^m - 1}{99} = \underbrace{1010\ldots10}_{m \text{ 10's}} \equiv 0 \pmod n$$

for any m divisible by $\varphi(99n)$. Take a very large such m, and then add on b distinct numbers of the form $10^{\varphi(n)r}$ for various even values of r; these all are $1 \pmod n$ and change some of the 1's to 3's. □

Now, we can solve the problem. Consider three cases:

- If $n = 2^k m$ where $\gcd(m, 10) = 1$ and $k \geq 2$ is even, then the concatenated number

$$10^k f\left(-\frac{g(k)}{10^k} \bmod m\right) + g(k)$$

works fine.

- If $n = 5^k m$ where $\gcd(m, 10) = 1$ and $k \geq 2$ is even, then the concatenated number

$$10^k f\left(-\frac{h(k)}{10^k} \bmod m\right) + h(k)$$

  works fine.

- If $n = 50m$ where $\gcd(m, 10) = 1$, then the concatenated number

$$100 f\left(-\frac{1}{2} \bmod m\right) + 50$$

  works fine.

Since every non-multiple of 20 divides such a number, we are done.